

Task 3 - IMS Android Prototype Codebase Documentation

Compliance Summary

- UI is implemented natively with Jetpack Compose.
- Dashboard + selected modules work end-to-end in the prototype.
- Flows and screen coverage are aligned with the high-fidelity designs from Task 2.
- Workflow intent matches the IMS system scope for this assignment prototype.

Tech Stack

- Kotlin
- Jetpack Compose (Material 3)
- Android Navigation Compose
- Room (SQLite)
- Coroutines / Flow
- Gradle (KTS)

App Identifier Requirement

- BuildConfig field: APP_IDENTIFIER
- Value: 2024101116.2024101015
- Defined in: app/build.gradle.kts

Module Coverage (Task 3 Prototype)

- Dashboard
 - Role-based home, stats, quick actions, navigation
- Attendance (end-to-end)
 - Mark attendance
 - My attendance
 - Attendance reports
- Examinations (end-to-end)
 - Exam list/filter
 - Create exam
 - Record marks/grades
 - Results view + reports

- Academy utilities
 - Manage courses/batches
 - Student categories
 - Graduation tracking
- Notifications
 - List, send, read/unread handling
- Profile & Settings
 - Edit profile
 - Sign out to login
 - App settings and identifier display

Codebase Map

- `app/src/main/java/com/ims/app/navigation/`
 - App routes and navigation graph
- `app/src/main/java/com/ims/app/ui/`
 - Compose screens by module (`dashboard`, `attendance`, `examination`, `academy`, `notification`)
- `app/src/main/java/com/ims/app/data/`
 - Room entities, DAOs, repositories, seed data
- `app/src/main/java/com/ims/app/i18n/`
 - Language handling and localization helpers

Build and Run

From Task3/:

- Build debug APK: `make` or `make build`
- Clean build outputs: `make clean`

Gradle fallback:

- `./gradlew -Dorg.gradle.java.home=/usr/lib/jvm/java-17-openjdk-amd64 assembleDebug`

Entry Points

- App init: `app/src/main/java/com/ims/app/IMSApplication.kt`
- Navigation root: `app/src/main/java/com/ims/app/navigation/NavGraph.kt`
- Database seed/init: `app/src/main/java/com/ims/app/data/seed/SeedData.kt`

Notes

- Prototype uses seeded data for realistic module flows.
- Category and dashboard list views are integrated with live Room-backed data.